JavaScript Data Types

What is a Data Type?

A **data type** tells JavaScript what kind of value a variable stores.

Example

```
let name = "Rokon"; // String
let age = 23;       // Number
```

Types of Data

JavaScript has **2 main types** of data.

1. Primitive Data Types
2. Non-Primitive (Reference) Data Types

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1. Primitive Data Types

Definition

A **Primitive Data Type** stores a **single value**. It is **immutable** (cannot be changed directly).

There are **7 Primitive Data Types** in JavaScript.

Data Type	Description	Example
String	Text	`"Hello"`
Number	Integer or Decimal	`10`, `3.14`
Boolean	True or False	`true`
Undefined	Variable declared but no value assigned	`undefined`
Null	Intentionally empty value	`null`
BigInt	Very large integer	`12345678901234567890n`
Symbol	Unique identifier	`Symbol("id")`

Examples

String

```
let name = "Rokon";
console.log(name);
```

Number

```
let age = 23;
let price = 99.99;
```

Boolean

```
let isStudent = true;
```

Undefined

```
let city;
console.log(city);
```

Null

```
let car = null;
```

BigInt

```
let bigNumber = 1234567890123456789012345678901234567890n;
```

Symbol

```
let id = Symbol("userId");
```

2. Non-Primitive (Reference) Data Types

Definition

A **Non-Primitive Data Type** can store **multiple values**.

These are called **Reference Types** because they are stored by reference (memory address).

Examples include:

- Object
- Array
- Function

Object

Objects store data in **key-value pairs**.

```
let student = {
    name: "Rokon",
    age: 23,
    country: "Bangladesh"
};

console.log(student.name);
```

Array

An array stores **multiple values** in a single variable.

```
let fruits = ["Apple", "Mango", "Orange"];

console.log(fruits[0]);
```

Function

A function is a reusable block of code.

```
function greet() {
    console.log("Hello!");
}

greet();
```

Primitive vs Non-Primitive

Feature	Primitive	Non-Primitive
Stores	Single Value	Multiple Values
Mutable	No (Immutable)	Yes (Mutable)
Stored By	Value	Reference
Size	Fixed	Dynamic
Examples	String, Number, Boolean	Object, Array, Function

Easy Example

Primitive

```
let a = 10;
let b = a;

b = 20;

console.log(a); // 10
console.log(b); // 20
```

Explanation:

- `b` gets a **copy** of `a`.
- Changing `b` does **not** change `a`.

Non-Primitive

```
let person1 = {
    name: "Rokon"
};

let person2 = person1;

person2.name = "Rahim";

console.log(person1.name); // Rahim
console.log(person2.name); // Rahim
```

Explanation:

- `person1` and `person2` point to the **same object**.
- Changing one also changes the other.

Easy Way to Remember

Primitive = Single Box

```
age = 23
```

Each variable has its **own copy**.

Non-Primitive = 📍 Address

```
person ------\
               ---> { name: "Rokon" }
friend ------/
```

Both variables point to the **same object** in memory.

Interview Question

Q: What is the difference between Primitive and Non-Primitive Data Types?

Answer:

- Primitive data types store a **single value** and are copied by **value**.
- Non-primitive data types store **multiple values** and are copied by **reference**.
- Primitive values are immutable, while non-primitive values are mutable.